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USF Springboard Bootcamp AI/ML

Capstone Research

Recent advances in computer vision have significantly improved the feasibility of performing real time object detection on aerial imagery. Drone based object detection, however, remains a challenging problem due to factors such as small object size, camera motion, variable altitude, and changing viewpoints. As a result, much of the existing research has focused on improving detection accuracy and inference speed under these constraints, with comparatively less emphasis on downstream analytics, reasoning, and user interaction.

One line of research investigates the direct application of modern single stage detectors to drone imagery. Recent work applying YOLOv8 to drone based scenarios demonstrates that the model is capable of achieving strong performance for common aerial object classes such as people and vehicles while maintaining real time inference speeds (Al-Darraji et al., 2025). These studies establish YOLOv8 as a viable baseline for UAV based perception systems and motivate its selection as the core detection model in this capstone. However, this body of work primarily treats object detection as an end goal, with outputs limited to bounding boxes and confidence scores, and does not address how detection results can be transformed into higher-level, queryable information.

More advanced research explores architectural optimizations to YOLOv8 specifically tailored to UAV use cases. Modular YOLOv8 optimization approaches introduce changes to feature extraction layers and attention mechanisms in order to improve detection accuracy while reducing computational overhead (Bognár et al., 2024). Experimental results reported in this work demonstrate measurable gains over baseline YOLOv8 models in both precision and inference efficiency, particularly in time sensitive applications such as maritime rescue and aerial surveillance. While these optimizations provide valuable insights into performance tuning, they introduce additional architectural complexity and remain tightly coupled to specific datasets and tasks. Importantly, these systems still focus exclusively on detection performance and do not extend into semantic reasoning, event abstraction, or user facing analytics.

In parallel to academic research, several open source systems demonstrate how object detection outputs can be transformed into structured, persistent representations. OpenDataCam is a notable example of a YOLO based system that converts frame-level detections into structured event data, enabling object tracking, counting, and temporal aggregation (OpenDataCam Contributors, n.d.). Rather than storing raw video, OpenDataCam emphasizes metadata first design, persisting detection results as JSON records that can be queried and visualized. This architectural pattern highlights a critical insight for scalable video analytics: long-term value is derived not from the video itself, but from the structured events extracted from it. However, OpenDataCam is primarily designed for fixed camera installations and supports only limited forms of interaction, such as predefined counts and visual summaries, without semantic querying or reasoning.

Cloud native video analytics pipelines further reinforce this separation between perception and analytics. Dataflow based video analytics systems demonstrate how detection outputs can be streamed into databases and processed independently of the vision model (Google Cloud Platform, n.d.). These pipelines emphasize scalability, fault tolerance, and structured storage, but generally lack interactive querying mechanisms and are not designed to support natural language interfaces or agentic reasoning.

Industry systems provide additional evidence of the demand for semantic interaction with video derived data. Commercial “AI search” platforms allow users to retrieve video segments using natural language queries such as “show all vehicles between 2–4 PM” or “find instances of people entering restricted areas” (Rhombus Systems, 2024). Although proprietary, these systems validate the usability gap in traditional computer vision pipelines and demonstrate that detection accuracy alone is insufficient for real world investigative and monitoring tasks. The ability to query, summarize, and reason over detection metadata is increasingly becoming a core requirement.

More recent developments in agentic video systems introduce the idea of placing a reasoning or validation layer on top of computer vision outputs. Event review frameworks use higher-level models to interpret detection results, flag anomalies, and answer user questions about system outputs (NVIDIA, 2024). These approaches suggest a promising direction in which object detection serves as a perceptual foundation, while an agentic layer provides interpretability, validation, and user interaction.

Based on this survey, several consistent limitations emerge across existing work. Most systems stop at producing detections or simple aggregates, offer limited support for exploratory or semantic queries, and lack mechanisms for reasoning over uncertainty or validating ambiguous detections. Few projects integrate object detection, structured event storage, and natural language interaction into a unified pipeline.

To address these gaps, this capstone proposes an end-to-end system that combines a YOLOv8-based drone object detection model with a structured event storage layer and an agentic natural language querying interface. Detection outputs are transformed into machine readable event records containing object class, temporal information, spatial metadata, and confidence scores. An AI agent operates over this structured data to answer user queries, generate summaries, and support exploratory analysis of drone footage. This design prioritizes usability, interpretability, and extensibility over raw detection performance alone.

As part of this effort, publicly available implementations are selected for reproduction to establish performance baselines and validate architectural decisions. Specifically, a modular YOLOv8 optimization implementation is reproduced to understand performance trade-offs in aerial detection (Bognár et al., 2024), while an event-based system such as OpenDataCam is reproduced to evaluate structured metadata pipelines (OpenDataCam Contributors, n.d.). Insights gained from these reproductions directly inform the design choices and expected performance of the proposed capstone system.

References

Al-Darraji, S., Al-Mansoori, A., & Hussein, M. (2025). *Real-time object detection using YOLOv8 model: A drone-based approach*. Journal of Wireless and Ubiquitous Applications, 16(1).
https://jowua.com/wp-content/uploads/2025/04/2025.I1.011.pdf

Bognár, B., Kovács, L., & Tóth, Z. (2024). *Modular YOLOv8 optimization for real-time UAV maritime rescue object detection*. Scientific Reports, Nature Publishing Group.
https://www.nature.com/articles/s41598-024-75807-1

Open Source Implementations and Public Codebases

Bognár, B. (2024). *modular_yolov8_sds: Official implementation of modular YOLOv8 optimization*. GitHub repository.
https://github.com/bgno1/modular_yolov8_sds

OpenDataCam Contributors. (n.d.). *OpenDataCam: Real-time object detection, tracking, and counting system*. GitHub repository.
https://github.com/opendatacam/opendatacam

Google Cloud Platform. (n.d.). *Dataflow video analytics pipeline*. GitHub repository.
https://github.com/GoogleCloudPlatform/dataflow-video-analytics

AWS Samples. (n.d.). *Hosting YOLOv8 on Amazon SageMaker endpoints*. GitHub repository.
https://github.com/aws-samples/host-yolov8-on-sagemaker-endpoint

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Cloud Infrastructure and Deployment References

Amazon Web Services. (2023). *Hosting YOLOv8 PyTorch models on Amazon SageMaker endpoints*. AWS Machine Learning Blog. https://aws.amazon.com/blogs/machine-learning/hosting-yolov8-pytorch-model-on-amazon-sagemaker-endpoints/

Amazon Web Services. (2022). *Scaling YOLOv5 inference with Amazon SageMaker endpoints and AWS Lambda*. AWS Machine Learning Blog. https://aws.amazon.com/blogs/machine-learning/scale-yolov5-inference-with-amazon-sagemaker-endpoints-and-aws-lambda/

Industry Systems and Agentic Video Analytics

Rhombus Systems. (2024). *Introducing AI search for video analytics*. https://www.rhombus.com/blog/introducing-ai-search/

NVIDIA. (2024). *Video search and summarization (VSS): Event reviewer documentation*. https://docs.nvidia.com/vss/2.4.1/content/vss_event_reviewer.html

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Datasets

Zhu, P., Wen, L., Du, D., & Bian, X. (2018). *VisDrone: The vision meets drone object detection benchmark*. https://github.com/VisDrone/VisDrone-Dataset

Du, D., Zhu, P., Wen, L., Bian, X., Ling, H., Hu, Q., Peng, T., & Zheng, J. (2018). *UAVDT: A large-scale benchmark for UAV-based object detection and tracking*. https://sites.google.com/site/daviddo0323/projects/uavdt